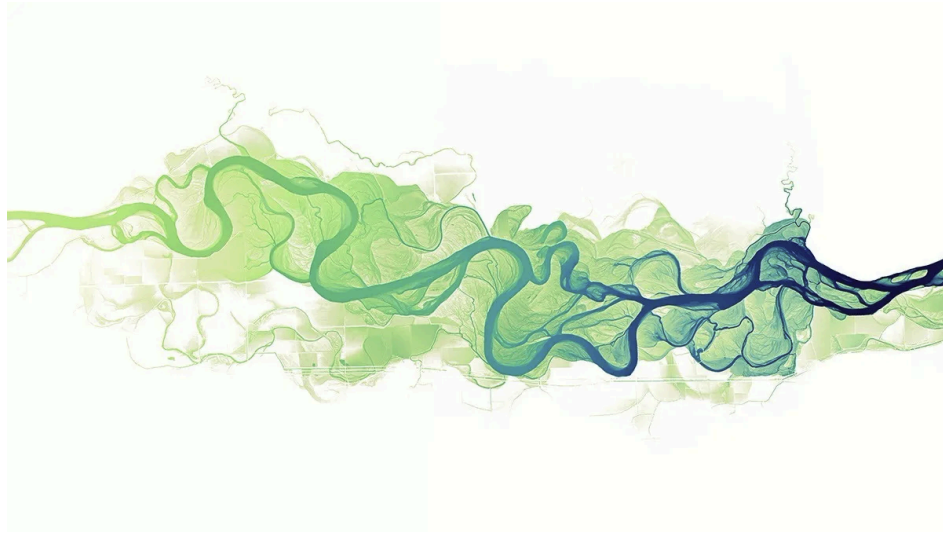


Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning



Bachelor Thesis

by
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Bachelors 2023

Submission date: 2026-07-02

Study Direction: Applied Digital Life Science

Supervisors:

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Imprint

Recommended Citation:

Florian Vollmer (2026). *Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning*

.

Keywords: Deep Learning, Object Detectors, Evaluation Metrics, Digital Lab and Production, Factory Production, Image Processing

Abstract

English Version

German Version

Acknowledgement

I would like to express my sincere gratitude to Dr. Stefan Glüge for setting up the project for this thesis and for his valuable support and guidance throughout the writing and compilation process of this thesis.

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List of Abbreviations

- **AI:** Artificial Intelligence
- **AP:** Average Precision
- **CNN:** Convolutional Neural Network
- **COCO:** Common Objects in Context
- **CV:** Cross Validation
- **FN:** False Negative
- **FP:** False Positive
- **IoU:** Intersection over Union
- **mAP:** Mean Average Precision
- **RF-DETR:** Roboflow Detection Transformer
- **TN:** True Negative
- **TP:** True Positive
- **YOLO:** You Only Look Once

1. Introduction

1.1. Context and Motivation

Systemic mastocytosis (SM) is a rare clonal hematopoietic neoplasm defined by the pathological expansion and accumulation of neoplastic mast cells (MC) in extracutaneous organs, most notably the bone marrow, but also the spleen, liver, and gastrointestinal tract. The major diagnostic criterion for SM is the presence of multifocal MC clusters in the bone marrow and/or extracutaneous organs, while minor criteria include an elevated serum tryptase level, MC expression of CD25, CD2, or CD30, and the presence of activating KIT mutations [1]. Despite these structured criteria, SM is frequently underdiagnosed in routine clinical practice. One of the central reasons for this is that mast cells, although morphologically distinct, occur only in very small numbers within a given bone marrow aspirate sample, making their manual identification and quantification an exhausting, subjective, and time-consuming task for hematopathologists.

The morphological assessment of mast cells carries direct diagnostic weight. The abnormal morphology of mast cells, defined as atypical spindle-shaped cells with hypogranulated cytoplasm and an oval nucleus, represents the first minor SM criterion. At least 25% of all MC must exhibit these morphological features in bone marrow smears or sections for this criterion to be fulfilled [1]. This quantitative threshold requires a reliable and reproducible count of both typical and atypical mast cells within a sample, a task that is very error-prone when performed manually and at scale.

In clinical practice, the integration of clinical signs and symptoms together with bone marrow morphological, immunophenotypic, and molecular results is required to diagnose SM variants [2]. However, the morphological step remains the entry point into this diagnostic chain. Standardizing and automating this step through computational means therefore has direct implications for diagnostic accuracy and clinical throughput. The University Hospital Zurich (USZ), which operates as the clinical partner for this thesis, has a concrete need for such a tool that could be embedded in its existing cell classification infrastructure.

1.2. Research Questions

1.3. Overall Concept

1.4. State of the Art

2. Methodology

2.1. Biology

2.2. Technical

Data Acquisition & Annotation

Data Preprocessing

Model Selection & Architecture

Model Training & Finetuning Strategy

Model Testing

Evaluation Metrics

UI-Creation

Implementation in USZ Infrastructure

3. Results

3.1. Model Results and Statistical Evaluation

3.2. Quantitative and Qualitative Evaluation

3.3. Evaluation Visualization

3.4. Real World Usage

Tests With Unseen Data

4. Discussion

4.1. Interpretation of Results

4.2. Clinical Relevance

4.3. Error Analysis

5. Conclusion

5.1. Summary of Key Findings

5.2. Limitations and Challenges

5.3. Future Research Directions

6. Declaration on AI Assistance

The author declares that the empirical research, data analysis, experimental design, and quantitative results presented in this thesis were generated independently and strictly remain the intellectual property of the author.

Artificial intelligence (AI) assistance was utilized solely for the purpose of language refinement, stylistic editing, and optimization of academic reading flow. Specifically, the AI model (Gemini 2.5 Pro) was employed to translate text, correct grammar, and restructure sentences into the required third-person past tense and formal academic style.

No core scientific content, model selection decisions, hyperparameter choices, or interpretation of results were delegated to or generated by the AI model.

7. References

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8. Lists

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9. Attachments